

WATSON LAKE, YT, Canada

WMO: 711990

Lat: 60.1142N

Lon: 128.8237W

Elev: 683

StdP: 93.38

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.1	-40.0	-45.2	0.0	-42.2	-42.7	0.1	-37.9	7.3	-10.4	6.1	-10.3	0.3	280	0.707

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	26.2	14.9	24.0	13.9	22.1	13.1	15.8	23.8	14.9	21.9	14.1	20.4	3.0	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.1	10.2	17.1	12.1	9.5	16.4	11.2	9.0	15.9	46.6	24.1	43.9	21.9	41.5	20.5	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	6.9	5.9	-46.7	29.5	3.4	1.8	-49.2	30.8	-51.1	31.9	-53.0	32.9	-55.5	34.2		
			-44.5	18.1	4.3	1.7	-47.5	19.4	-50.0	20.4	-52.4	21.4	-55.5	22.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.1	-21.5	-17.4	-10.5	-0.1	8.0	13.5	15.3	13.4	7.7	-0.1	-13.9	-20.3
	DBStd	14.73	10.01	7.93	8.03	4.85	4.17	3.27	2.57	3.22	3.30	4.63	7.65	8.59
	HDD10.0	4834	976	767	635	304	89	4	1	6	82	313	718	939
	HDD18.3	7466	1234	1001	893	554	320	151	101	157	318	571	968	1198
	CDD10.0	424	0	0	0	0	28	109	163	111	13	0	0	0
	CDD18.3	14	0	0	0	0	1	6	6	3	0	0	0	0
	CDH23.3	256	0	0	0	0	11	77	98	69	1	0	0	0
	CDH26.7	45	0	0	0	0	1	17	16	11	0	0	0	0
Wind	WSAvg	1.8	0.8	1.0	1.9	2.4	2.5	2.4	2.3	2.2	2.2	2.1	1.2	0.9
Precipitation	PrecAvg	436	30	25	19	16	37	54	60	47	47	37	29	32
	PrecMax	589	53	72	46	46	118	108	140	94	98	78	58	66
	PrecMin	310	12	1	1	2	4	8	7	9	16	16	10	4
	PrecStd	70	10	16	12	12	24	26	31	21	17	15	14	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.7	3.5	8.0	14.9	24.7	29.0	28.6	28.3	21.1	12.8	5.5	2.6
		MCWB	0.9	0.9	3.1	6.6	12.4	15.3	15.9	16.3	13.3	8.0	2.6	1.2
	2%	DB	-0.9	0.1	5.5	11.6	21.7	25.5	26.0	25.3	17.2	9.9	2.2	-0.6
		MCWB	-2.0	-1.3	1.3	4.7	10.8	14.2	15.0	15.0	10.6	5.8	0.1	-1.8
	5%	DB	-5.3	-3.1	3.5	9.5	19.0	22.9	23.8	22.8	15.4	7.9	-0.6	-3.8
		MCWB	-6.0	-4.1	0.2	3.6	9.9	12.9	14.2	14.1	9.7	4.6	-1.8	-4.5
	10%	DB	-9.7	-5.8	1.6	7.6	16.4	20.7	21.9	20.5	13.8	6.2	-3.5	-8.8
		MCWB	-10.1	-6.5	-1.0	2.5	8.4	12.0	13.5	13.2	9.0	3.5	-4.2	-9.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	1.1	3.4	7.1	13.4	17.0	17.2	17.0	13.7	8.6	2.6	1.3
		MCDB	2.6	3.0	7.3	13.8	20.8	25.1	25.4	25.5	20.2	11.9	4.8	2.4
	2%	WB	-1.9	-1.4	1.5	5.2	11.5	15.1	15.8	15.7	11.7	6.6	0.2	-1.7
		MCDB	-0.7	0.0	4.8	10.7	19.8	23.1	23.4	23.5	15.6	9.3	1.9	-0.6
	5%	WB	-5.7	-4.1	0.4	3.9	10.2	13.8	15.0	14.7	10.5	4.9	-1.9	-4.5
		MCDB	-5.0	-3.1	3.3	8.8	18.1	21.2	21.6	21.3	14.0	7.4	-0.7	-3.7
	10%	WB	-9.9	-6.5	-0.9	2.8	9.1	12.8	14.2	13.7	9.5	3.6	-4.2	-9.3
		MCDB	-9.5	-5.8	1.5	6.9	15.7	19.4	20.3	19.3	13.0	5.8	-3.4	-8.9
Mean Daily Temperature Range	5% DB	MDBR	8.7	12.2	14.7	12.4	12.0	11.0	10.8	11.0	9.6	7.0	8.1	8.5
		MCDBR	11.8	12.8	14.3	14.5	16.3	14.9	15.1	15.7	12.7	9.7	9.8	12.1
		MCWBR	11.0	11.6	10.7	8.7	7.8	6.4	6.6	7.3	7.2	6.3	8.3	11.2
	5% WB	MCDBR	11.8	12.2	13.7	13.1	14.7	13.1	13.1	13.5	10.9	9.1	9.3	12.0
		MCWBR	11.1	11.1	10.4	8.1	7.4	6.0	6.1	6.8	6.6	6.2	7.9	11.1
Clear-Sky Solar Irradiance	taub	0.213	0.240	0.263	0.326	0.353	0.369	0.369	0.355	0.312	0.273	0.219	0.186	
	taud	2.164	2.319	2.289	2.173	2.363	2.380	2.401	2.460	2.566	2.551	2.196	2.021	
	Ebn at Noon	599	799	882	868	873	861	852	838	828	760	579	464	
	Edh at Noon	39	61	88	120	108	108	104	91	69	50	37	28	
All-Sky Solar Radiation	RadAvg	0.39	1.22	2.83	4.51	5.28	5.45	5.04	4.05	2.54	1.22	0.46	0.21	
	RadStd	0.04	0.13	0.23	0.29	0.42	0.33	0.36	0.25	0.23	0.11	0.06	0.02	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page